Topic: Complex Eigenvalues

Question: Solve the system of differential equations.

$$\vec{x}' = \begin{bmatrix} -1 & 2 \\ -5 & 1 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 2 \\ 1+3i \end{bmatrix} e^{3it} + c_2 \begin{bmatrix} 2 \\ 1-3i \end{bmatrix} e^{-3it}$$

B $$\vec{x} = c_1 \begin{bmatrix} 2 \\ 1+3i \end{bmatrix} e^{-3it} + c_2 \begin{bmatrix} 2 \\ 1-3i \end{bmatrix} e^{3it}$$

C $$\vec{x} = c_1 \begin{bmatrix} 2 \\ 3i \end{bmatrix} e^{3it} + c_2 \begin{bmatrix} 2 \\ -3i \end{bmatrix} e^{-3it}$$

D $$\vec{x} = c_1 \begin{bmatrix} 2 \\ 3i \end{bmatrix} e^{-3it} + c_2 \begin{bmatrix} 2 \\ -3i \end{bmatrix} e^{3it}$$

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Solution: A

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} -1 & 2 \\ -5 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} -1 & 2 \\ -5 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} -1 - \lambda & 2 \\ -5 & 1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (-1 - \lambda)(1 - \lambda) - (2)(-5)$$

$$|A - \lambda I| = \lambda^2 + 9$$

Solve the characteristic equation to find the Eigenvalues.

$$\lambda^2 + 9 = 0$$

$$\lambda = \pm 3i$$

Then for these complex conjugate Eigenvalues, $\lambda_1 = 3i$ and $\lambda_2 = -3i$, we find

$$A - (3i)I = \begin{bmatrix} -1 - 3i & 2 \\ -5 & 1 - 3i \end{bmatrix}$$

and

$$A - (-3i)I = \begin{bmatrix} -1 + 3i & 2 \\ -5 & 1 + 3i \end{bmatrix}$$

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If we turn these matrices back into systems of equations, we get

$$(-1-3i)k_1+2k_2=0 \qquad (-1+3i)k_1+2k_2=0$$

$$-5k_1+(1-3i)k_2=0 \qquad -5k_1+(1+3i)k_2=0$$

From the first equation in the first system for $\lambda_1=3i$, we find

$$k_2=\left(\frac{1}{2}+\frac{3}{2}i\right)k_1$$

Choosing $k_1=2$ gives $k_2=1+3i$, which leads us to the Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}2\\1+3i\end{bmatrix}$$

and the corresponding solution vector

$$\overrightarrow{x_1}=\overrightarrow{k_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_1}=\begin{bmatrix}2\\1+3i\end{bmatrix}e^{3it}$$

In the same way, from the first equation in the second system for $\lambda_2=-3i$, we find

$$k_2=\left(\frac{1}{2}-\frac{3}{2}i\right)k_1$$

Choosing $k_1=2$ gives $k_2=1-3i$, which leads us to the Eigenvector

$$\overrightarrow{k_2}=\begin{bmatrix}2\\1-3i\end{bmatrix}$$

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and the corresponding solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 2 \\ 1-3i \end{bmatrix} e^{-3it}$$

Therefore, the general solution will be

$$\overrightarrow{x} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 2 \\ 1+3i \end{bmatrix} e^{3it} + c_2 \begin{bmatrix} 2 \\ 1-3i \end{bmatrix} e^{-3it}$$

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Topic: Complex Eigenvalues

Question: Solve the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 1+i \\ 1 \end{bmatrix} e^{(2-i)t} + c_2 \begin{bmatrix} 1-i \\ 1 \end{bmatrix} e^{(2+i)t}$$

B $$\vec{x} = c_1 \begin{bmatrix} 1+i \\ 1 \end{bmatrix} e^{(2+i)t} + c_2 \begin{bmatrix} 1-i \\ 1 \end{bmatrix} e^{(2-i)t}$$

C $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1+i \end{bmatrix} e^{(2-i)t} + c_2 \begin{bmatrix} 1 \\ 1-i \end{bmatrix} e^{(2+i)t}$$

D $$\vec{x} = c_1 \begin{bmatrix} 1 \\ 1+i \end{bmatrix} e^{(2+i)t} + c_2 \begin{bmatrix} 1 \\ 1-i \end{bmatrix} e^{(2-i)t}$$

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Solution: B

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 3 - \lambda & -2 \\ 1 & 1 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (3 - \lambda)(1 - \lambda) - (-2)(1)$$

$$|A - \lambda I| = \lambda^2 - 4\lambda + 5$$

Solve the characteristic equation to find the Eigenvalues.

$$\lambda^2 - 4\lambda + 5 = 0$$

$$(\lambda - 2)^2 = -1$$

$$\lambda = 2 \pm i$$

Then for these complex conjugate Eigenvalues, $\lambda_1 = 2 + i$ and $\lambda_2 = 2 - i$, we find

$$A - (2 + i)I = \begin{bmatrix} 3 - (2 + i) & -2 \\ 1 & 1 - (2 + i) \end{bmatrix}$$

$$A - (2 + i)I = \begin{bmatrix} 1 - i & -2 \\ 1 & -1 - i \end{bmatrix}$$

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and

$$A-(2-i)I=\begin{bmatrix}3-(2-i) & -2\\ 1 & 1-(2-i)\end{bmatrix}$$

$$A-(2-i)I=\begin{bmatrix}1+i & -2\\ 1 & -1+i\end{bmatrix}$$

If we turn these matrices back into systems of equations, we get

$$(1-i)k_1-2k_2=0 \qquad (1+i)k_1-2k_2=0$$

$$k_1+(-1-i)k_2=0 \qquad k_1+(-1+i)k_2=0$$

From the second equation in the first system for $\lambda_1=2+i$, we find

$$k_1=(1+i)k_2$$

Choosing $k_2=1$ gives $k_1=1+i$, which leads us to the Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}1+i\\ 1\end{bmatrix}$$

and the corresponding solution vector

$$\overrightarrow{x_1}=\overrightarrow{k_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_1}=\begin{bmatrix}1+i\\ 1\end{bmatrix}e^{2+it}$$

In the same way, from the second equation in the second system for $\lambda_2=2-i$, we find

$$k_1=(1-i)k_2$$

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Choosing $k_2 = 1$ gives $k_1 = 1 - i$, which leads us to the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 1-i \\ 1 \end{bmatrix}$$

and the corresponding solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 1-i \\ 1 \end{bmatrix} e^{2-it}$$

Therefore, the general solution will be

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 1+i \\ 1 \end{bmatrix} e^{(2+i)t} + c_2 \begin{bmatrix} 1-i \\ 1 \end{bmatrix} e^{(2-i)t}$$

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Topic: Complex Eigenvalues

Question: Solve the system of differential equations.

$$\vec{x}' = \begin{bmatrix} 2 & 4 \\ -5 & 6 \end{bmatrix} \vec{x}$$

Answer choices:

A $$\vec{x} = c_1 \begin{bmatrix} 1+2i \\ 2 \end{bmatrix} e^{(4-4i)t} + c_2 \begin{bmatrix} 1-2i \\ 2 \end{bmatrix} e^{(4+4i)t}$$

B $$\vec{x} = c_1 \begin{bmatrix} 1+2i \\ 2 \end{bmatrix} e^{(4+4i)t} + c_2 \begin{bmatrix} 1-2i \\ 2 \end{bmatrix} e^{(4-4i)t}$$

C $$\vec{x} = c_1 \begin{bmatrix} 2 \\ 1+2i \end{bmatrix} e^{(4-4i)t} + c_2 \begin{bmatrix} 2 \\ 1-2i \end{bmatrix} e^{(4+4i)t}$$

D $$\vec{x} = c_1 \begin{bmatrix} 2 \\ 1+2i \end{bmatrix} e^{(4+4i)t} + c_2 \begin{bmatrix} 2 \\ 1-2i \end{bmatrix} e^{(4-4i)t}$$

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Solution: D

We'll need to find the determinant $|A - \lambda I|$,

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 4 \\ -5 & 6 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right|$$

$$|A - \lambda I| = \left| \begin{bmatrix} 2 & 4 \\ -5 & 6 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right|$$

$$|A - \lambda I| = \begin{vmatrix} 2 - \lambda & 4 \\ -5 & 6 - \lambda \end{vmatrix}$$

$$|A - \lambda I| = (2 - \lambda)(6 - \lambda) - (4)(-5)$$

$$|A - \lambda I| = \lambda^2 - 8\lambda + 32$$

Solve the characteristic equation to find the Eigenvalues.

$$\lambda^2 - 8\lambda + 32 = 0$$

$$(\lambda - 4)^2 = -16$$

$$\lambda = 4 \pm 4i$$

Then for these complex conjugate Eigenvalues, $\lambda_1 = 4 + 4i$ and $\lambda_2 = 4 - 4i$, we find

$$A - (4 + 4i)I = \begin{bmatrix} 2 - (4 + 4i) & 4 \\ -5 & 6 - (4 + 4i) \end{bmatrix}$$

$$A - (4 + 4i)I = \begin{bmatrix} -2 - 4i & 4 \\ -5 & 2 - 4i \end{bmatrix}$$

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and

$$A-(4-4i)I=\begin{bmatrix}2-(4-4i) & 4\\ -5 & 6-(4-4i)\end{bmatrix}$$

$$A-(4-4i)I=\begin{bmatrix}-2+4i & 4\\ -5 & 2+4i\end{bmatrix}$$

If we turn these matrices back into systems of equations, we get

$$(-2-4i)k_1+4k_2=0 \qquad (-2+4i)k_1+4k_2=0$$

$$-5k_1+(2-4i)k_2=0 \qquad -5k_1+(2+4i)k_2=0$$

From the first equation in the first system for $\lambda_1=4+4i$, we find

$$k_2=\left(\frac{1}{2}+i\right)k_1$$

Choosing $k_1=2$ gives $k_2=1+2i$, which leads us to the Eigenvector

$$\overrightarrow{k_1}=\begin{bmatrix}2\\ 1+2i\end{bmatrix}$$

and the corresponding solution vector

$$\overrightarrow{x_1}=\overrightarrow{k_1}e^{\lambda_1 t}$$

$$\overrightarrow{x_1}=\begin{bmatrix}2\\ 1+2i\end{bmatrix}e^{(4+4i)t}$$

In the same way, from the first equation in the second system for $\lambda_2=4-4i$, we find

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$$k_2 = \left(\frac{1}{2} - i\right) k_1$$

Choosing $k_1 = 2$ gives $k_2 = 1 - 2i$, which leads us to the Eigenvector

$$\overrightarrow{k_2} = \begin{bmatrix} 2 \\ 1 - 2i \end{bmatrix}$$

and the corresponding solution vector

$$\overrightarrow{x_2} = \overrightarrow{k_2} e^{\lambda_2 t}$$

$$\overrightarrow{x_2} = \begin{bmatrix} 2 \\ 1 - 2i \end{bmatrix} e^{(4-4i)t}$$

Therefore, the general solution will be

$$\overrightarrow{x} = c_1 \overrightarrow{x_1} + c_2 \overrightarrow{x_2}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 2 \\ 1 + 2i \end{bmatrix} e^{(4+4i)t} + c_2 \begin{bmatrix} 2 \\ 1 - 2i \end{bmatrix} e^{(4-4i)t}$$

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